



Create offset profile

Type an offset profile as regions or points -- width and height against station -- and get exact wires for Driven edge offset and Driven offset surface.

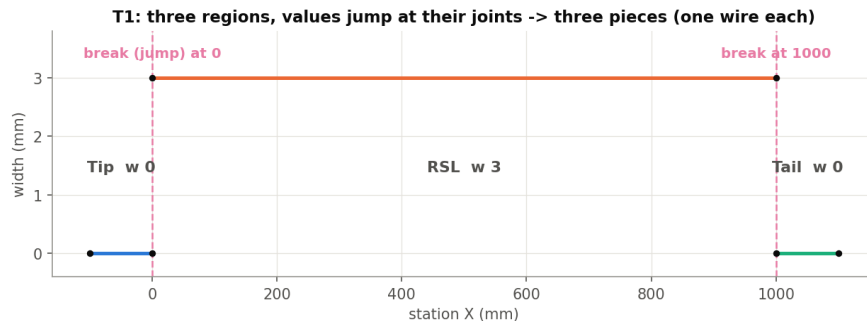


What it does

Driven edge offset (DEO) and Driven offset surface (DOS) read an offset profile: wires in profile coordinates, X = station, Y = width offset, Z = height offset. Create offset profile builds those wires from a table, exactly (polynomial Beziers, no sampling), instead of a hand-drawn sketch.

Use it when

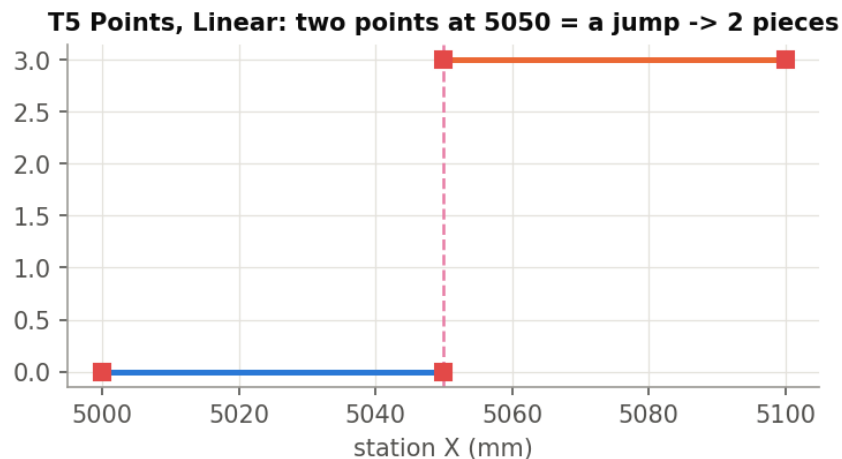
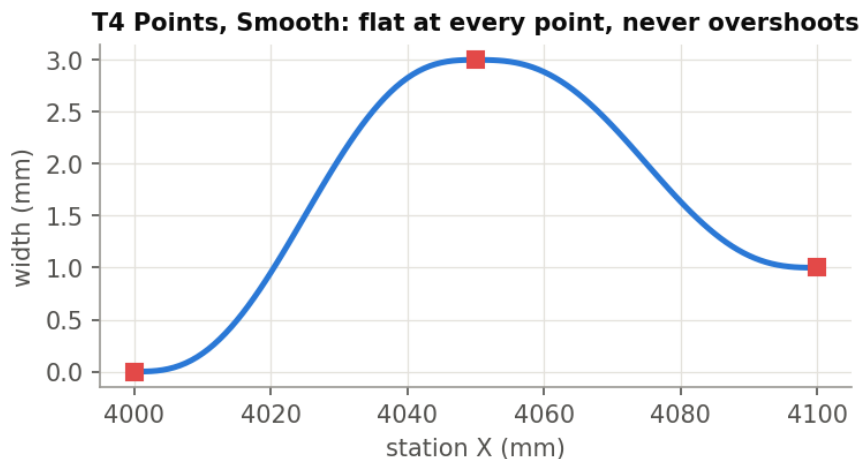
- The offset is defined by regions: tip, running length, tail ...
- Width or height ramps, holds or steps along X
- Stations come from the part: a vertex or mate connector



Test T1: Tip w 0, RSL w 3, Tail w 0. The values jump at 0 and 1000, so the profile breaks there: three wires.



Pieces and breaks



A piece = one wire

A continuous stretch of the profile, e.g. T4 left: three Smooth points, one piece. Touching regions with equal values join too.

A break

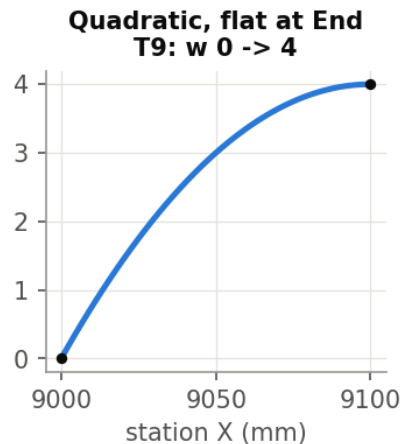
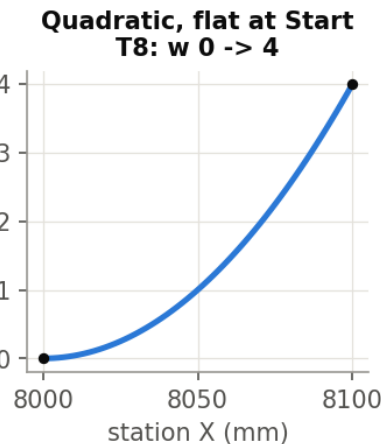
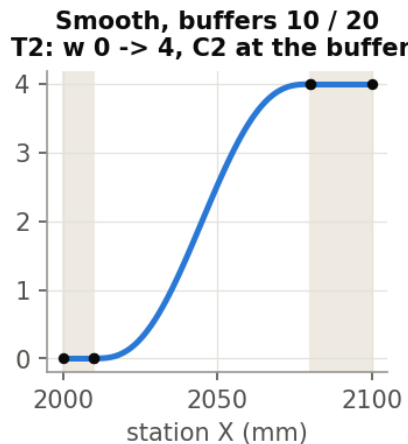
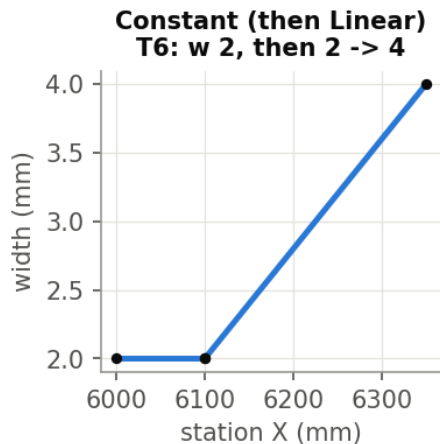
Where the offset jumps (touching regions or two points at one station, different values -- T5 right) or is undefined (a gap in X), the profile ends and a new piece starts.

Ends

The profile starts at the first item and stops at the last; nothing is extrapolated.



Region shapes



Constant / Linear

One width and height; or a straight change from start to end values (corners at the ends).

Smooth

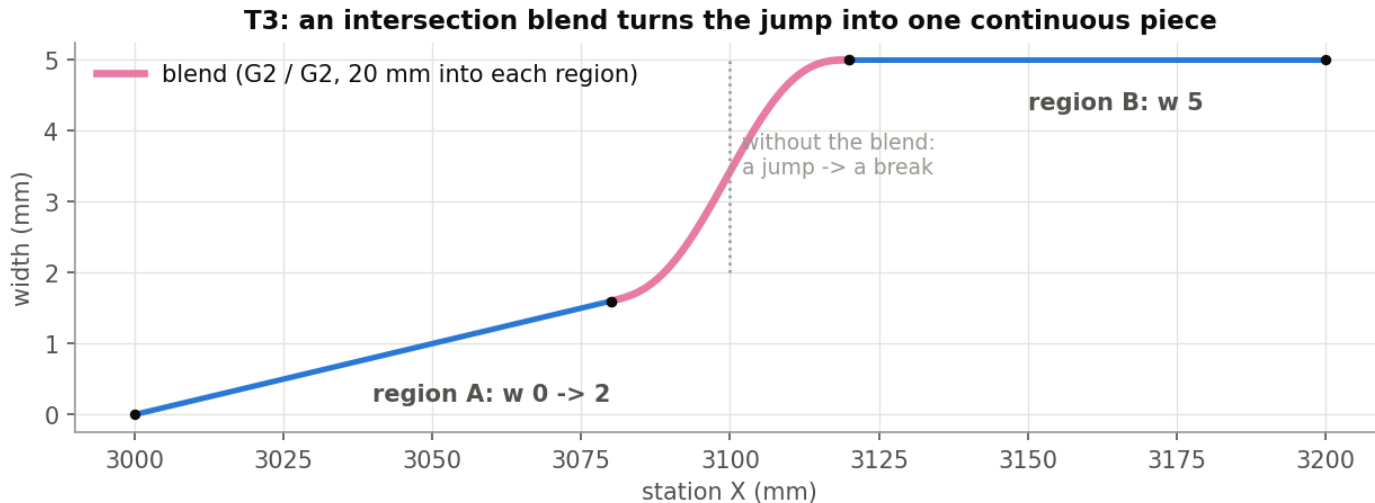
Smootherstep: flat, zero curvature at both ends of the change, never overshoots. Buffers hold the end values first (shaded).

Quadratic

A parabola flat at one end ('Flat at' Start or End), a kink at the other. Width and height use the same shape.



Blending two regions



Intersections

One entry per pair of consecutive regions, kept by region names. Blend off: join as they are if the values match, else break.

Blend on

A Hermite bridge from a distance into the first region to a distance into the second, G0 / G1 / G2 at each side.

Drag arrows

Each blend distance also has a drag arrow on the profile while the dialog is open.



The dialog: Regions (1/2)

T6 Regions constant 600...  

1 **Regions** Points

Regions  **CLEAR**

 Flat 

 Ramp 

2 Name Ramp

3 Enter X **At point**

4 Start point
T6 sketch point at x 6...  

5 Start offset along X 0 mm

#	Parameter	What it means
1	Mode	Regions or Points.
2	Region name	Empty = Region n. Intersections are kept by these names, so they must differ.
3	Start station	Enter X (type the station) or At point.
4	Start point	At point: a vertex or mate connector (one can be created in the field).
5	Start offset along X	Station = the point's world X plus this signed distance.



The dialog: Regions (2/2)

6 Enter X **At point**

End point
T6 mate connector at... ×

End offset along X 50 mm

7 Constant **Linear** Smooth Quadratic

8 Start width 2 mm

9 Intersections

Add Intersection

> Debug

Final

#	Parameter	What it means
6	End station	The same for the end. A region may be entered toward -X (start > end).
7	Shape	Constant, Linear, Smooth or Quadratic.
8	Start / end width, height	The values at each end (Constant: one Width, one Height).
9	Intersections	Between consecutive regions, rebuilt automatically (next slides).



The dialog: Quadratic and buffers

T8 Regions QUADRATIC ... ✓ ✕

Regions Points

Regions CLEAR

▼ Quad ✕

Name Quad

Enter X At point

Start X 8000 mm

Enter X At point

End X 8100 mm

Constant Linear Smooth Quadratic

Start End

Start width 0 mm

End width 4 mm

Start height 0 mm

End height 0 mm

Start buffer 0 mm

Intersections

Add Intersection

> Debug

Final

#	Parameter	What it means
1	Shape	Quadratic (test T8).
2	Flat at	Start or End: the end of the change with zero slope; the other end meets its neighbour at a kink.
3	Start / end width, height	End values; width and height follow the same shape.
4	Start / end buffer	Hold the start (end) value this far in from that end; the change runs between the buffers. Not for Constant.

Entered toward -X, each end keeps its own values and buffer, and a quadratic's flat end stays at the end it was given (T11).



The dialog: an intersection blend (1/2)

T3 Regions G2 / G2 blend... ☒ ☐

Regions Points

Regions CLEAR

> A

> B

Add Region

1 Intersections CLEAR

▼ A / B

First region A

Second region B

2 ☒ Blend

3 Continuity with first region

Curvature

#	Parameter	What it means
1	Intersections	One per consecutive pair (here A / B), labelled by the region names.
2	Blend	Off: touching regions with equal values join, anything else breaks. On: a blend.
3	Continuity with first region	G0, G1 (tangent) or G2 (curvature) where the blend leaves region A.



The dialog: an intersection blend (2/2)

4 Distance into first region 20...

5 Continuity with second region

Curvature ▼

6 Distance into second r... 2...

Add Intersection

> Debug

Final

#	Parameter	What it means
4	Distance into first region	How far back from A's end the blend starts.
5	Continuity with second region	The same where it joins region B.
6	Distance into second region	How far past B's start it ends. Both 0 only works across a gap in X.



The dialog: Points

1 T4 Points smooth 0 / 3 / ... ☒ ☐

Regions **Points**

Points

2 > 4000 mm

3 ✓ 4050 mm

4 **Enter X** At point

5 X 4050 mm

4 Width 3 mm

5 Height 2 mm

6 To next point Smooth

> 4100 mm

Add Point

> Debug

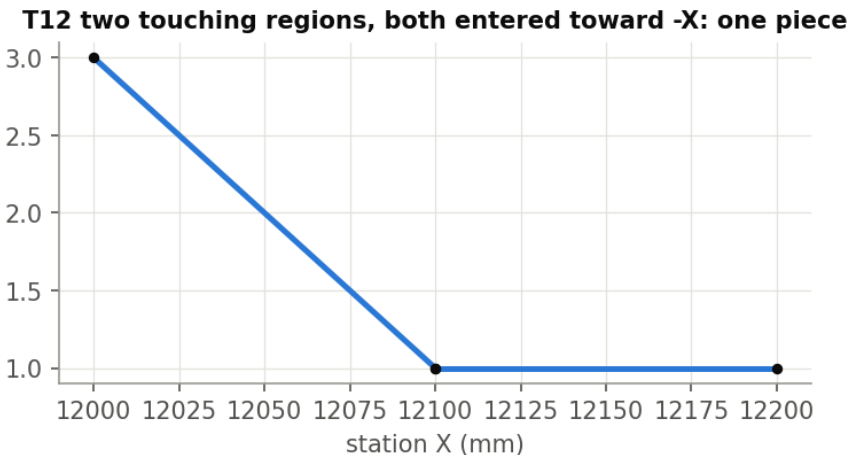
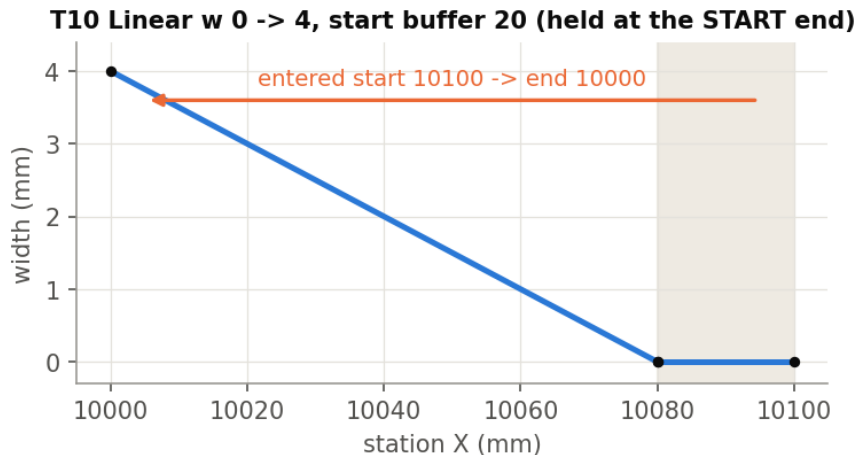
Final

#	Parameter	What it means
1	Mode	Points: the profile through a list of stations.
2	Station	Enter X or At point (vertex / mate connector plus a signed offset along +X).
3	X	The station, typed.
4	Width	Width offset at this station.
5	Height	Height offset at this station.
6	To next point	Smooth (flat at both points, no overshoot), Linear, or Hold (keep this value to the next point, then break). Ignored on the last point.

Two points at one station with different values are a jump: the profile breaks there (T5).



Example: regions entered toward -X (T10, T12)



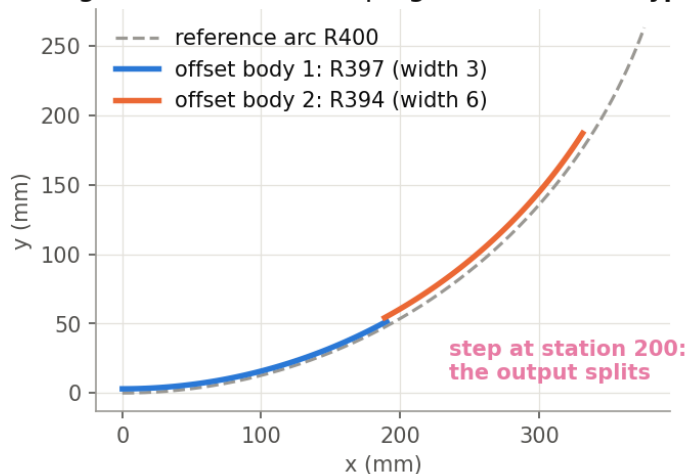
For a part whose tip is at +X, enter regions from the tip: start 10100, end 10000. The same region stated low to high.

T10: Linear w 0 -> 4 with a start buffer of 20: width 0 held from 10100 to 10080, 4 at 10000. T12: two touching regions both entered toward -X join into one piece (w 1 at 12150, 2 at 12050).



Driven edge offset takes regions directly

DEO regions test D3: regions Low 0..200 w 3 | High 200..400 w 6, typed in Driven edge offset



Profile source: Regions

Driven edge offset has the same Regions table (same parameters, from offset_profile_core); no wire is made. Tests D1-D5: identical to the wire, within 1e-3 mm.

Stations

A region's X means what a wire's X means under DEO's Measure along. 'At point' stations land exactly on the part only when measuring along World X.

Not built yet

Per-break None / Line join in DEO / DOS. Today every break splits the output (D3: two offset bodies at the step).



Outputs and messages

Published for Extract variables

Key	What it holds
output	All piece wires (T1: 3).
piece1 .. pieceN	Each piece, in station order.
startVertex / endVertex	Where the profile starts (lowest station) / ends (highest).
breakVertices	The piece ends on both sides of every break (T1: 4).
pieceCount, breakCount	Numbers (T1: 3 and 2).
breakStations	Station of each break: a jump, or the middle of a gap (T1: 0, 1000 mm).
startStation, endStation	First and last station (T1: -100, 1100 mm).

Messages

	Message / what to do
Error	Add at least one region. / Add at least two points at different stations.
Error	Regions 'A' and 'B' overlap. Regions may touch, not overlap.
Error	Region 'x': the buffers are as long as the region or longer.
Error	Two regions are named 'x' ... Intersections are kept by name.
Error	The blend between 'A' and 'B' has no length but the offset jumps ... Give a distance, or leave a gap in X.
Error	Start point: pick a point (vertex or mate connector) for the station. At point with nothing picked.



Tips and limits



Name regions by role

Tip, RSL, Tail ... Intersection settings follow the names when regions are reordered.



Smooth for joins that must not kink

A buffer into a Smooth ramp is curvature-continuous (C2); a Linear corner is only G0.



Print segments

Debug group: every piece and sub-segment with its stations, degree and end values.



Breaks split the output

DEO / DOS have no per-break None / Line choice yet: a step in the profile is a step (a gap) in the offset.



One shape per region

Width and height share the region's shape.



Stations are world X

Picked points give their world X; on a part measured along its reference, place stations accordingly.



Where it lives

Document	driven_offset / BeamBuilder Testbed (f61d2c000ab2d1240776342e)
Tabs	create_offset_profile (the feature), offset_profile_core (regions / points -> exact pieces; shared with Driven edge offset)
Tests	Offset profile tests studio: T1-T12 (+ 3 error cases) -- devtools/onshape/build_offset_profile_tests.py, check_offset_profile.py; DEO regions tests: D1-D5 (5/5 pass 2026-09-29)
Design	driven_offset/research_create_offset_profile.md; driven_offset/DOCUMENT_MAP.md
Related	Driven edge offset, Driven offset surface (consumers); Evaluate offset (a profile measured from existing edges)